

PHARMACOVIGILANCE INTERNSHIP – TASK 1

ADR Identification Basics

Prepared by: RISHITA MAURYA
Fictional educational training case

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Prepared by: Pharmacovigilance Intern

Case type: Fictional educational training case

Purpose: Demonstrate foundational understanding of ADR identification and basic pharmacovigilance concepts.

1. Introduction to Pharmacovigilance

Pharmacovigilance (PV) is the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or other medicine-related problems.

Role of Pharmacovigilance in Public Health

Pharmacovigilance helps to:

- Protect patients from avoidable medicine-related harm.
- Detect previously unknown or insufficiently characterized safety problems.
- Monitor the safety of medicines after they are used in larger and more diverse populations.
- Identify safety signals and support further evaluation.
- Support a favourable benefit–risk balance of medicines.
- Provide information that can contribute to regulatory and public-health action.

Key principle: Clinical trials provide important safety information, but uncommon, delayed, population-specific or otherwise difficult-to-detect reactions may become clearer after wider use of a medicine. Continuous safety monitoring is therefore important.

2. Adverse Event (AE) vs Adverse Drug Reaction (ADR)

Feature	Adverse Event (AE)	Adverse Drug Reaction (ADR)
Definition	Any untoward medical occurrence in a patient receiving a pharmaceutical product; a causal relationship does not have to be established.	A noxious, unintended response to a medicine occurring at normally used doses.
Causality	Does not necessarily imply that the medicine caused the event.	Implies a suspected/assessed relationship with the medicine.
Example	A patient taking an antibiotic develops a headache; the cause has not yet been assessed.	A patient develops an urticarial rash shortly after starting a medicine, with improvement after withdrawal and no better explanation identified.

Simple distinction:

AE = What happened?

ADR = What happened and is the medicine suspected to be responsible?

3. Types of ADRs

3.1 Predictable vs Unpredictable ADRs

Type A – Augmented / Predictable

- Usually related to the known pharmacological action of the medicine.
- Often dose-related.
- Generally more predictable from the pharmacology of the medicine.
- Example: **Hypoglycaemia caused by insulin**, particularly when the insulin dose exceeds the patient's immediate glucose requirement.

Type B – Bizarre / Unpredictable

- Not usually related to the medicine's usual pharmacological action.
- Often difficult to predict.
- May include allergic or idiosyncratic reactions.
- Example: **Anaphylactic reaction to penicillin** in a susceptible patient.

3.2 Classical Type A–F Classification

Type	Name	Basic idea	Example
A	Augmented	Dose-related, predictable from pharmacology	Hypoglycaemia with insulin
B	Bizarre	Unpredictable/idiosyncratic or hypersensitivity-type	Penicillin allergy
C	Chronic	Associated with long-term use	Adrenal suppression with prolonged corticosteroid use
D	Delayed	Appears after some delay	Certain drug-related cancers or teratogenic effects
E	End-of-use	Associated with withdrawal	Withdrawal symptoms after abrupt discontinuation of certain medicines
F	Failure	Unexpected failure of therapy	Reduced contraceptive effectiveness due to an interacting medicine

4. ADR Identification Scenario (Fictional Scenario)

Case

A 42-year-old woman with hypertension and seasonal allergic rhinitis is taking amlodipine 5 mg once daily for approximately one year and cetirizine 10 mg once daily for two months. She starts **amoxicillin 500 mg three times daily** for a bacterial sinus infection.

On **Day 3**, she develops a widespread red, itchy rash over the chest, back and arms with mild facial swelling. She reports no new foods, cosmetics, insect bites or other newly started medicines. Amoxicillin is discontinued. The rash and itching begin to decrease the next day, and by Day 6 the rash is almost completely resolved and the facial swelling has disappeared. Amoxicillin is not restarted.

Assessment

- **Adverse event/reaction:** Widespread pruritic rash with mild facial swelling.
- **Possible medicines:** Amoxicillin, cetirizine and amlodipine should initially be reviewed.
- **Most likely suspect:** Amoxicillin.
- **Reason:** It was newly started immediately before the reaction, while the other medicines were long-standing therapies.
- **Temporal relationship:** Reaction occurred approximately three days after amoxicillin initiation.
- **Dechallenge:** Positive — symptoms improved after amoxicillin was withdrawn.
- **Rechallenge:** Not performed.
- **Alternative causes:** No obvious new food, cosmetic, insect-bite or newly started medicine exposure was reported.
- **Preliminary conclusion:** The event is consistent with a potential amoxicillin-associated hypersensitivity/allergic-type ADR. A temporal relationship and positive dechallenge support the preliminary assessment but do not, by themselves, establish definitive causality.

5. Teratogenic Drugs and Pregnancy Risk Categories

Historical FDA Pregnancy Letter Categories: A, B, C, D and X

The A–X letter system is important historically and for understanding older references, but the **FDA removed these letter categories from prescription drug labeling in 2015** because they were considered overly simplistic. Current FDA labeling uses narrative pregnancy risk information instead.

Category	Historical basic meaning	Simple educational example
A	Adequate, well-controlled studies in pregnant women did not demonstrate fetal risk.	Some prenatal vitamins/components when appropriately used
B	Animal studies did not demonstrate fetal risk, but adequate human studies were lacking, or animal findings were not confirmed in adequate human studies.	Amoxicillin was historically classified as Category B
C	Animal studies showed an adverse fetal effect and adequate human studies were lacking, or there were insufficient data; use required benefit–risk consideration.	Fluconazole was historically assigned Category C for certain uses/doses

D	Evidence of human fetal risk exists, but potential benefits may warrant use in some circumstances.	Phenytoin was historically Category D
X	Evidence of fetal abnormalities/risk outweighed any possible benefit; use in pregnancy was contraindicated.	Isotretinoin was historically Category X

Important PV point

Teratogenicity refers to the potential of an exposure to cause developmental abnormalities or birth defects. Pregnancy safety assessment should not rely only on a letter category. Current labeling requires a more detailed description of available human/animal data, risks and clinical considerations.

6. Causality Assessment

Causality assessment is the evaluation of how likely it is that a medicine caused an observed adverse reaction.

Three commonly discussed approaches are:

6.1 Bradford Hill Criteria

The Bradford Hill considerations are a framework for evaluating whether an observed association may represent a causal relationship, particularly in epidemiology and population-level evidence.

Common considerations include:

- Strength of association
- Consistency
- Specificity
- Temporality
- Biological gradient
- Plausibility
- Coherence
- Experiment
- Analogy

Example

If multiple independent studies repeatedly show that a medicine is associated with a particular adverse outcome, the association has a plausible biological mechanism, and the exposure consistently precedes the outcome, several Bradford Hill considerations may support a causal interpretation.

Important: Bradford Hill considerations are not a simple patient-level scoring algorithm like the Naranjo questionnaire.

6.2 WHO-UMC Causality Assessment Scale

The **WHO-UMC system** classifies the strength of the relationship between a medicine and an adverse event using clinical assessment of factors such as:

- Reasonable time relationship between drug exposure and event
- Alternative explanations
- Response after withdrawal (dechallenge)

- Response after re-exposure (rechallenge), when available
- Known pharmacology or previous information about the reaction

Common categories include:

- Certain
- Probable/Likely
- Possible
- Unlikely
- Conditional/Unclassified
- Unassessable/Unclassifiable

Example

A patient develops angioedema after using an ACE inhibitor, improves after the medicine is withdrawn, and no better alternative explanation is identified. The case may support a **Probable/Likely** relationship under WHO-UMC assessment, depending on the complete clinical information.

6.3 Naranjo Algorithm

The **Naranjo Algorithm** is a structured questionnaire used to estimate the probability that an adverse event is related to a medicine.

It considers questions such as:

- Did the reaction occur after the medicine was administered?
- Did the reaction improve after withdrawal?
- Did the reaction recur after reintroduction?
- Are there alternative causes?
- Is there objective evidence supporting the reaction?
- Was the reaction previously reported?

A numerical score is obtained and interpreted using the Naranjo categories.

Example

If a patient develops a rash after starting a medicine, the rash improves after withdrawal, there is no convincing alternative cause, and the reaction has previously been reported with the medicine, the Naranjo score may support a **probable** relationship depending on the answers to all questionnaire items.

Important: A Naranjo score should be calculated from the actual case information; it should not be guessed from the final diagnosis.

7. How to Report an ADR in India

Suspected ADR Reporting Form

Healthcare professionals can use the **Suspected ADR Reporting Form** under PvPI.

Important information includes:

1. Patient initials and demographic information

2. Date of reaction onset
3. Description of the reaction
4. Management and outcome
5. Relevant medical and medication history
6. Seriousness criteria, where applicable
7. Suspected medicine(s), including dose, route, frequency and therapy dates
8. Action taken with the suspected medicine
9. Rechallenge information, if applicable
10. Concomitant medicines and other relevant products
11. Relevant investigations
12. Reporter details

Basic reporting workflow

Identify suspected reaction → collect patient/reaction/drug details → complete the ADR form → submit to the appropriate ADR Monitoring Centre (AMC)/PvPI channel → information is assessed and contributes to medicine-safety monitoring.

Healthcare professionals and consumers can report suspected ADRs. PvPI encourages reporting of suspected reactions, including known or unknown and serious or non-serious reactions.

8. Overview of the Pharmacovigilance Programme of India (PvPI)

The **Pharmacovigilance Programme of India (PvPI)** is the Government of India's flagship drug-safety monitoring programme.

Key functions

- Collect adverse event/ADR reports.
- Collate and analyze safety information.
- Monitor medicine-related safety issues.
- Support identification and evaluation of safety signals.
- Improve patient safety and welfare.
- Communicate safety information to relevant stakeholders.
- Provide recommendations to the **Central Drugs Standard Control Organization (CDSCO)** for appropriate regulatory action.

The **National Coordination Centre (NCC) for PvPI** is located at the **Indian Pharmacopoeia Commission (IPC), Ghaziabad**.

PvPI operates through **ADR Monitoring Centres (AMCs)** and other reporting channels. Reports contribute to systematic medicine-safety monitoring and can support regulatory decision-making.

9. Public Health Importance of ADR Reporting

ADR reporting is important because individual reports can contribute to a larger body of safety evidence.

A simplified pharmacovigilance pathway is:

Patient/Healthcare Professional → ADR Report → AMC/PvPI → Assessment & Data Analysis → Signal Detection/Evaluation → Regulatory/Public-Health Action → Improved Patient Safety

The objective is not merely to collect forms. The purpose is to transform safety information into knowledge that can reduce medicine-related harm.

10. Key Takeaways

1. **AE** describes an untoward medical occurrence and does not necessarily imply causality.
2. **ADR** describes a noxious, unintended response associated with a medicine at normally used doses.
3. Type A reactions are generally predictable and pharmacologically related; Type B reactions are generally unpredictable and may be idiosyncratic or hypersensitivity-related.
4. The A–X pregnancy letter categories are a **historical FDA classification** and are no longer used in current FDA prescription labeling.
5. Bradford Hill considerations help evaluate causal relationships at an epidemiological/population level.
6. WHO-UMC provides a structured clinical approach to causality categories.
7. The Naranjo Algorithm uses a questionnaire and score to estimate the probability of a drug-event relationship.
8. Suspected ADRs can be reported even when causality is not established.
9. PvPI is India's national drug-safety monitoring programme and works through ADR Monitoring Centres and other reporting channels.
10. Good pharmacovigilance depends on accurate, timely and sufficiently detailed safety information.

Declaration: The ADR identification scenario is fictional and does not represent a real patient.